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Configuring SSH Remote Access on Ubuntu Server

Objective

Set up an Ubuntu Server with SSH (Secure Shell) remote access and connect to it from a host desktop machine using Bridged Network mode.

Operating System Installation

Install Ubuntu Server

1. Start the VM and boot from the Ubuntu Server ISO
2. Follow the installation wizard:
 - Profile setup: Create a user account (e.g., server / server123)
 - Install OpenSSH server: **Check this option** (Important!)
 - Selected software: **None** (Standard Ubuntu Server)
3. Wait for installation to complete and reboot

Network Configuration

Configure Network Mode

For VirtualBox:

1. Power off the VM (if running)
2. Go to **Settings** → **Network**
3. **Adapter 1**: Enable Network Adapter
4. **Attached to**: Select **Bridged Adapter**
5. **Name**: Select your host's physical network adapter (Wi-Fi/Ethernet)
6. Click **OK**

For VMware:

1. Power off the VM
2. Go to **VM Settings** → **Network Adapter**

3. Select **Bridged** mode

4. Click **OK**

Install/Verify SSH Server

Server: Install SSH on the server (if not already installed):

- `sudo systemctl status ssh` – for checking the SSH status.
- `sudo apt update` – updating the your server.
- `sudo apt install openssh-server -y` –installing the SSH.
- `sudo systemctl enable ssh` – enabling the SSH to your unit. (SERVER/DESKTOP)
- `sudo systemctl start ssh` – starting the SSH to your unit. (SERVER/DESKTOP)

Bridged makes the server behave like a separate PC in your LAN.

1. Find the Server's IP

Inside the VM, run:

```
Sample@UBUNTU-DESKTOP:~$ ip a
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
        valid_lft forever preferred_lft forever
2: enp0s3: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    link/ether 08:00:27:9b:8c:ca brd ff:ff:ff:ff:ff:ff
    inet 192.168.1.4/24 brd 192.168.1.255 scope global dynamic noprefixroute enp0s3
        valid_lft 83602sec preferred_lft 83602sec
    inet6 2001:4451:b00:2000:d76c:1af2:a10c:f36d/64 scope global temporary dynamic
        valid_lft 179068sec preferred_lft 83386sec
    inet6 2001:4451:b00:2000:a00:27ff:fe9b:8cca/64 scope global dynamic mngtmpadr proto kernel_r
        valid_lft 179068sec preferred_lft 92668sec
    inet6 fe80::a00:27ff:fe9b:8cca/64 scope link proto kernel_ll
        valid_lft forever preferred_lft forever
Sample@UBUNTU-DESKTOP:~$
```

From the desktop, connect to the server: `ssh username@192.168.1.120`

2. Connect from Desktop

From your host desktop:

```
Sample@UBUNTU-DESKTOP:~$ ssh Server@192.168.1.14
Server@192.168.1.14's password:
Welcome to Ubuntu 24.04.1 LTS (GNU/Linux 6.8.0-79-generic x86_64)

 * Documentation:  https://help.ubuntu.com
 * Management:    https://landscape.canonical.com
 * Support:       https://ubuntu.com/pro

System information as of Mon Sep  8 11:24:38 PM UTC 2025

System load:            0.0
Usage of /:             5.1% of 48.91GB
Memory usage:          5%
Swap usage:            0%
Processes:             139
Users logged in:       1
IPv4 address for enp0s3: 192.168.1.14
IPv6 address for enp0s3: 2001:4451:b00:2000:a00:27ff:fe57:bc4a

Expanded Security Maintenance for Applications is not enabled.

136 updates can be applied immediately.
To see these additional updates run: apt list --upgradable

Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status

Last login: Mon Sep  8 22:52:07 2025 from 192.168.1.4
Server@UBUNTU-SERVER:~$
```

Submission Requirements

1. Screenshots (labeled clearly):

- VM network settings showing Bridged Adapter
- Server IP address (output of `ip a` or `hostname -I`)
- SSH service status (`sudo systemctl status ssh`)
- Successful SSH connection from host
- Verification commands executed inside VM

2. Submission Format:

- Single PDF file
- Include name, student ID, and date
- All screenshots with brief explanation
- Upload to your portfolio website.

UBUNTU SERVER SSH REMOTE ACCESS LAB

Name: Aspacio Gian Paul J.

Student ID: 202200357D

Course/Year/Section: BSIT 3B

Date Submitted: _____

Instructor: _____

1 OBJECTIVES (5 POINTS)

Check all that apply:

- ☐ Create and configure a Virtual Machine with Ubuntu Server
- ☐ Configure Bridged Network Adapter for network access
- ☐ Determine and verify the server's IP address
- ☐ Install OpenSSH Server and enable the service
- ☐ Establish successful SSH connection from host machine
- ☐ Execute verification commands to confirm remote access

2 HARDWARE & SOFTWARE REQUIREMENTS (5 POINTS)

Item	Specification
Host Operating System	Linux
Virtualization Software (VirtualBox/VMware)	Oracle
Ubuntu Server Version	Ubuntu 64bit
VM Name	vmgian
RAM Allocated	8000MB
Storage Allocated	1000 GB
Network Mode	Intel Pro
SSH Client Used	Secure Shell Client

3 VM CONFIGURATION

Setting	Value
VM Name	UBUNTU-SERVER
OS Type	Ubuntu (64-bit)
Memory	8000 MB
Hard Disk	100 GB
Network Adapter	Bridged

4. NETWORK CONFIGURATION

Commands Executed: ssh gian@10.0.2.4

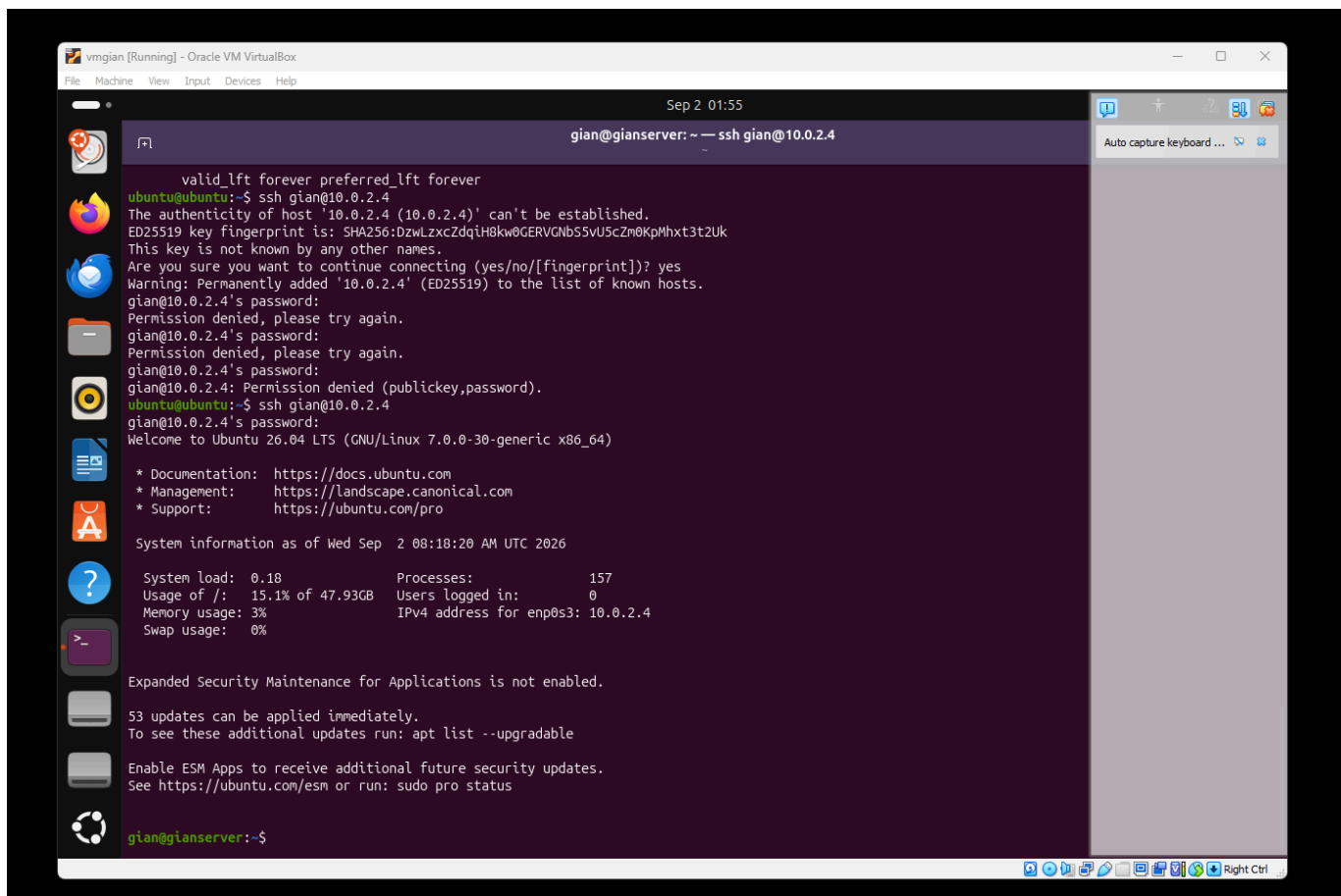
Server IP Address: 10.0.2.4

Network Interface: NAT Network

6. REMOTE CONNECTION

Connection Details:

Parameter	Value
Server IP	10.0.2.4
Username	gian
SSH Port	22



```
vmgian [Running] - Oracle VM VirtualBox
File Machine View Input Devices Help

Sep 2 01:55
gian@gianserver: ~ -- ssh gian@10.0.2.4

valid_lft forever preferred_lft forever
ubuntu@ubuntu:~$ ssh gian@10.0.2.4
The authenticity of host '10.0.2.4 (10.0.2.4)' can't be established.
ED25519 key fingerprint is: SHA256:DZwLzxcZdqIH0kw0GERVGNb55vU5cZm0KpMhxt3t2Uk
This key is not known by any other names.
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added '10.0.2.4' (ED25519) to the list of known hosts.
gian@10.0.2.4's password:
Permission denied, please try again.
gian@10.0.2.4's password:
Permission denied, please try again.
gian@10.0.2.4's password:
gian@10.0.2.4: Permission denied (publickey,password).
ubuntu@ubuntu:~$ ssh gian@10.0.2.4
gian@10.0.2.4's password:
Welcome to Ubuntu 26.04 LTS (GNU/Linux 7.0.0-30-generic x86_64)

 * Documentation:  https://docs.ubuntu.com
 * Management:    https://landscape.canonical.com
 * Support:       https://ubuntu.com/pro

System information as of Wed Sep  2 08:18:20 AM UTC 2026

System load:  0.18           Processes:    157
Usage of /:   15.1% of 47.93GB Users logged in: 0
Memory usage: 3%            IPv4 address for enp0s3: 10.0.2.4
Swap usage:   0%

Expanded Security Maintenance for Applications is not enabled.

53 updates can be applied immediately.
To see these additional updates run: apt list --upgradable

Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status

gian@gianserver:~$
```

```
vm_ko [Running] - Oracle VM VirtualBox
File Machine View Input Devices Help

Ubuntu 26.04 LTS gianserver tty1

gianserver login: gian
Password:
Welcome to Ubuntu 26.04 LTS (GNU/Linux 7.0.0-30-generic x86_64)

 * Documentation:  https://docs.ubuntu.com
 * Management:    https://landscape.canonical.com
 * Support:       https://ubuntu.com/pro

System information as of Wed Sep  2 08:17:04 AM UTC 2026

System load:  0.63           Processes:            155
Usage of /:   15.1% of 47.9GB Users logged in:          0
Memory usage: 4%           IPv4 address for enp0s3: 10.0.2.4
Swap usage:   0%

Expanded Security Maintenance for Applications is not enabled.

53 updates can be applied immediately.
To see these additional updates run: apt list --upgradable

Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status

gian@gianserver:~$ ip a
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
        valid_lft forever preferred_lft forever
2: enp0s3: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc pfifo_fast state UP group default qlen 1000
    link/ether 08:00:27:82:3e:86 brd ff:ff:ff:ff:ff:ff
    altname enx080027823e86
    inet 10.0.2.4/24 metric 100 brd 10.0.2.255 scope global dynamic enp0s3
        valid_lft 593sec preferred_lft 593sec
    inet6 fe80::a00:27ff:fe82:3e86/64 scope link proto kernel_l1
        valid_lft forever preferred_lft forever
gian@gianserver:~$ _
```